OFFICIAL UI / UX REPORT

Z LAB -- Redesign & Visual Audit

Complete Desktop & Mobile Interface Specification

- Design System: MindMate Soft Pastel Claymorphic Pop (#F4ECF8 canvas, rounded-3xl cards, zero harsh borders)
- Signature Accents: Lime-to-pink gradient pills (#D7F75B to #FCA5BA), circular pastel badges, segmented dial gauges
- Part I: Desktop Workspace Interface (1440 x 900) -- Full hero, dashboard, 6 hubs, omni-search, and flagship tools
- Part II: Mobile Island Experience (390 x 844) -- Floating bottom nav island, touch drawers, and thumb-friendly controls
- Privacy Architecture: 100% Client-Side In-Browser RAM execution via Web Crypto, Canvas, Web Audio, and WebAssembly
- Coverage: 32+ High-Resolution Verified Screen States across all 6 core laboratory categories and 44 client utilities

Generated: Monday, October 12, 2026 - Next.js 16.4.0 Engine

PART I

DESKTOP WORKSPACE INTERFACE

1440 x 900 Viewport - High-DPI Desktop Layout

DESKTOP VIEW (1440 x 900)

Route: /

01. Home Screen -- Hero & Version 2.0 Launchpad

Z Lab | Version 2.0 | All Hubs | Studio | PDF Lab | Words | Study | Search ⌘K | 100% Local

Version 2.0

Z Lab



Unlock the Power of Your Device

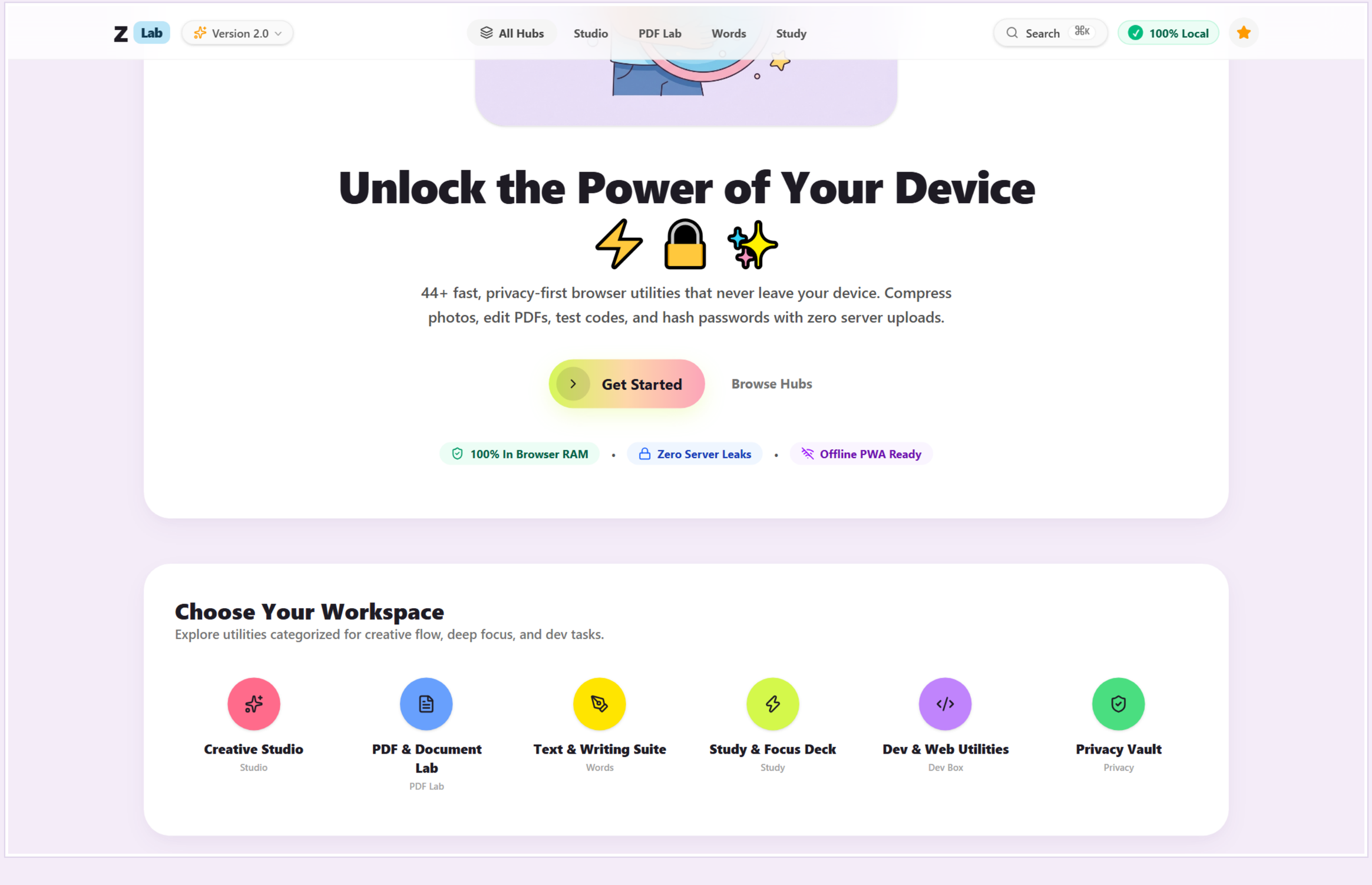
44+ fast, privacy-first browser utilities that never leave your device. Compress photos, edit PDFs, test codes, and hash passwords with zero server uploads.

Get Started | Browse Hubs

DESKTOP VIEW (1440 x 900)

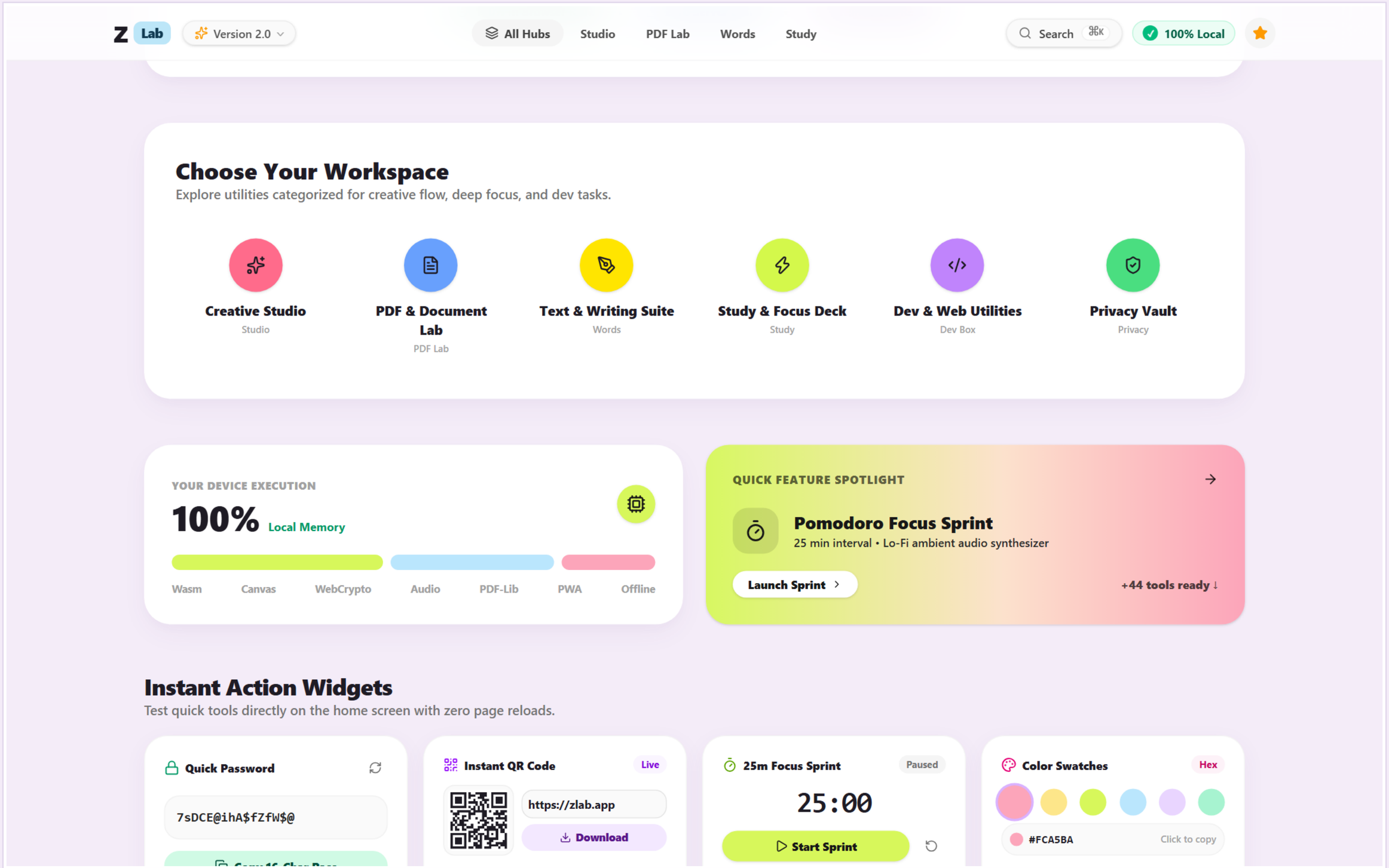
Route: /

02. Home Screen -- Dashboard, RAM Meter & Featured Spotlight



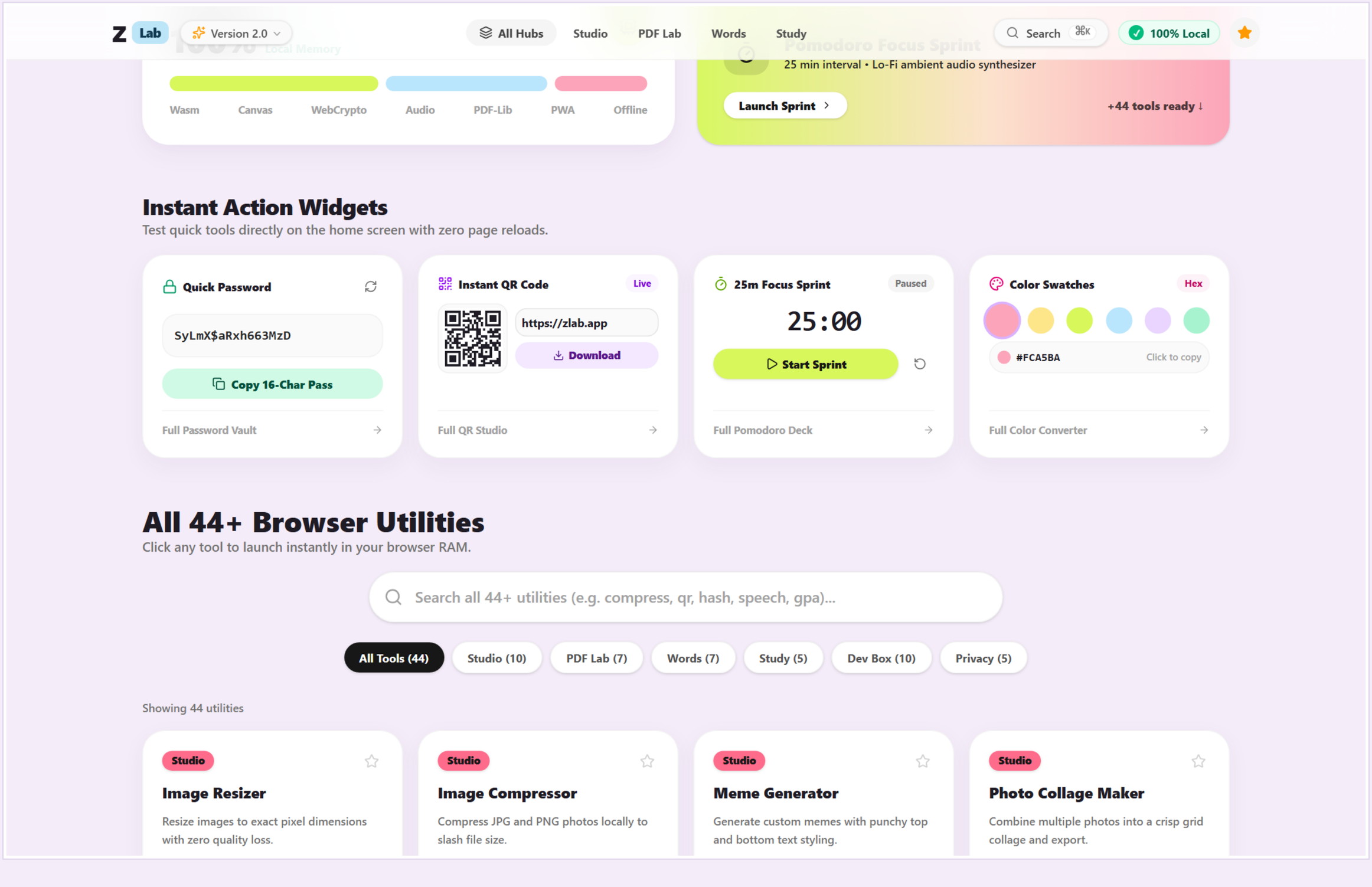
DESKTOP VIEW (1440 x 900) | Route: /

03. Home Screen -- Instant Action Bento Launchpad



DESKTOP VIEW (1440 x 900) | Route: /

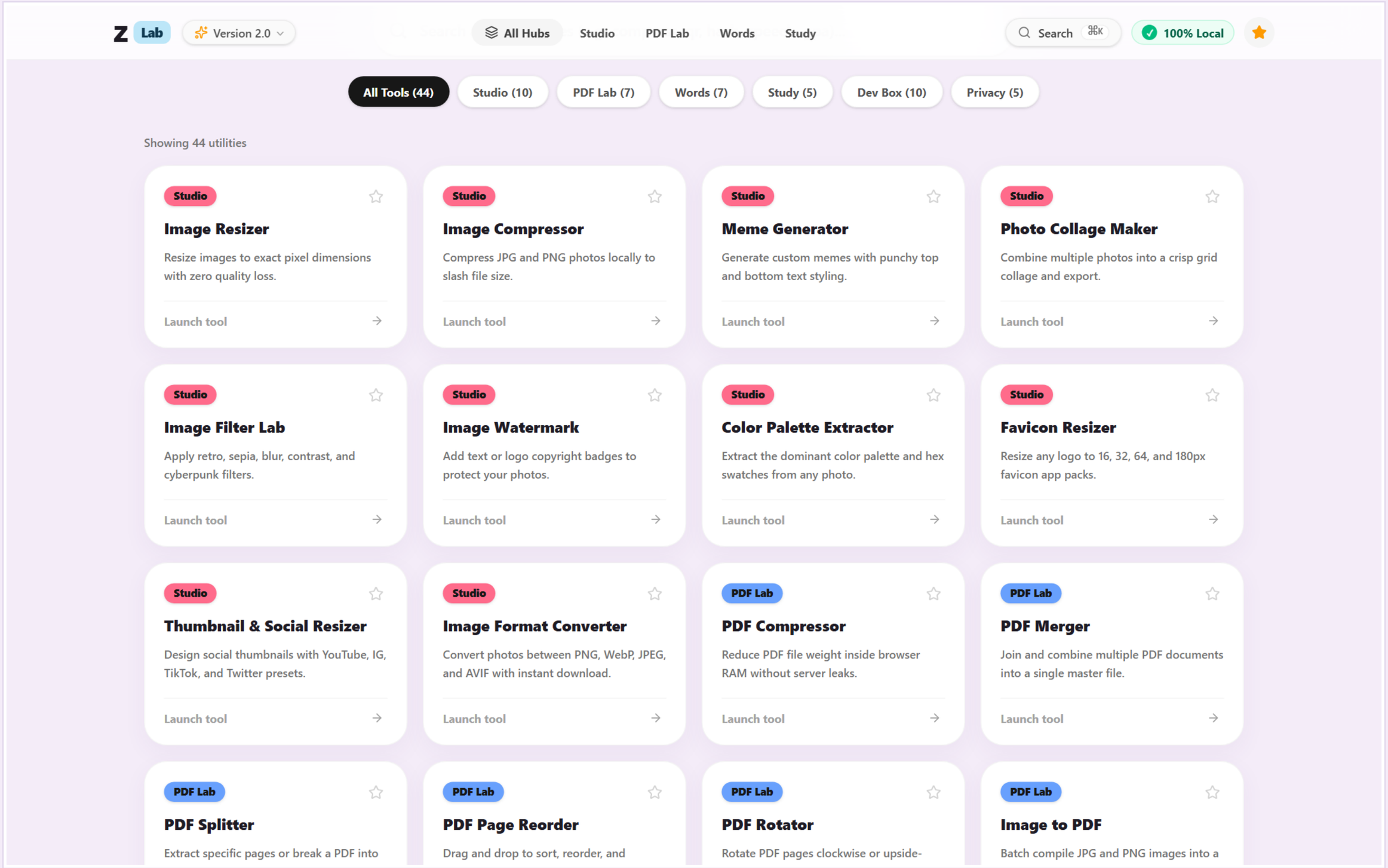
04. Home Screen -- 6 Dedicated Laboratory Hubs Showcase



DESKTOP VIEW (1440 x 900)

Route: /

05. Home Screen -- Tools Explorer & Category Pill Filters



DESKTOP VIEW (1440 x 900)

Route: /

06. Global Omni-Search & Command Palette (Ctrl+K)

Z Lab Version 2.0 All Hubs Studio PDF Lab Words Study Search ⌘K 100% Local

Version 2.0

Z Lab



Unlock the Power of Your Device

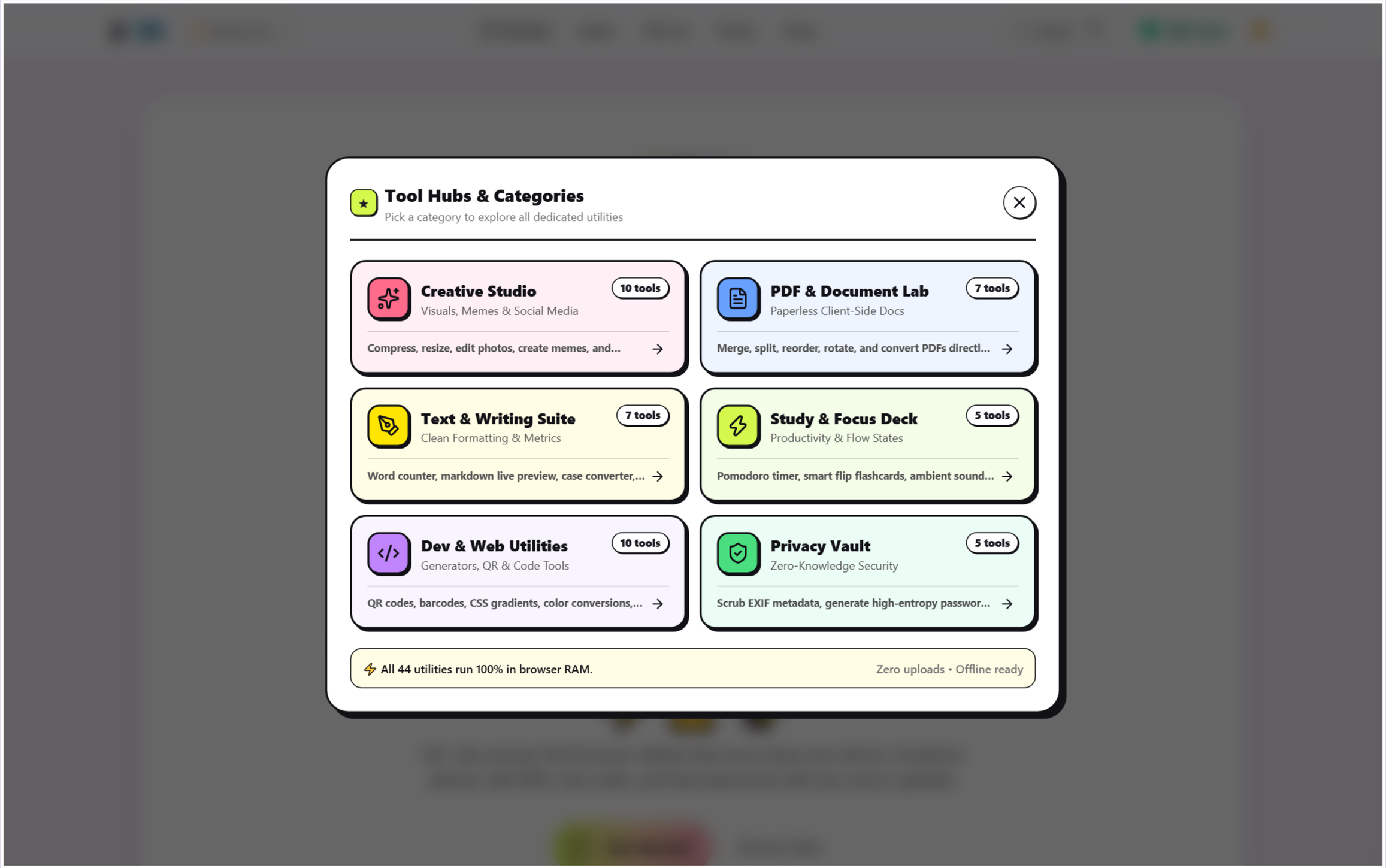
44+ fast, privacy-first browser utilities that never leave your device. Compress photos, edit PDFs, test codes, and hash passwords with zero server uploads.

Get Started Browse Hubs

DESKTOP VIEW (1440 x 900)

Route: /

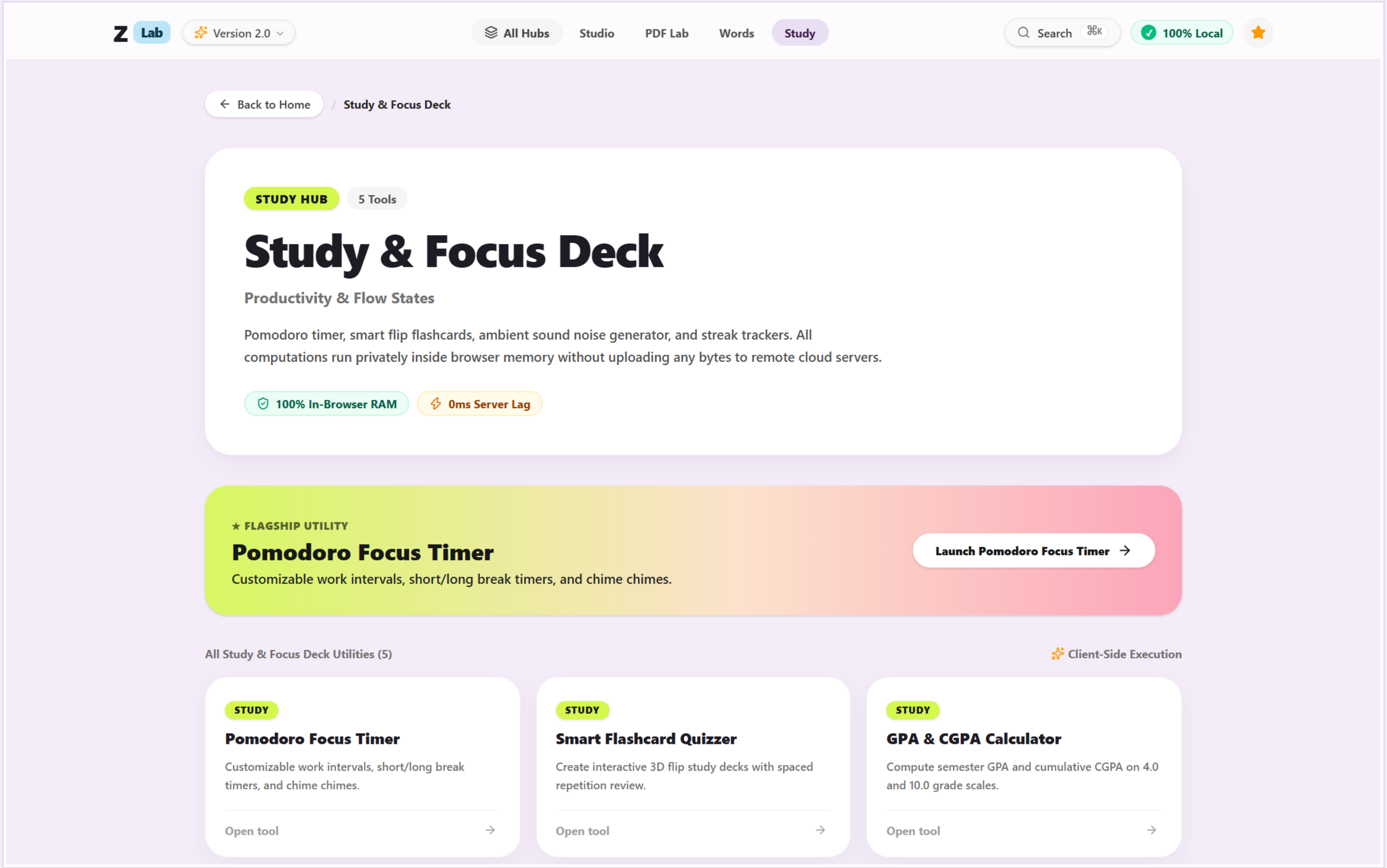
07. Global Navigation Drawer -- Category Hubs Directory



DESKTOP VIEW (1440 x 900)

Route: /tools/study-focus

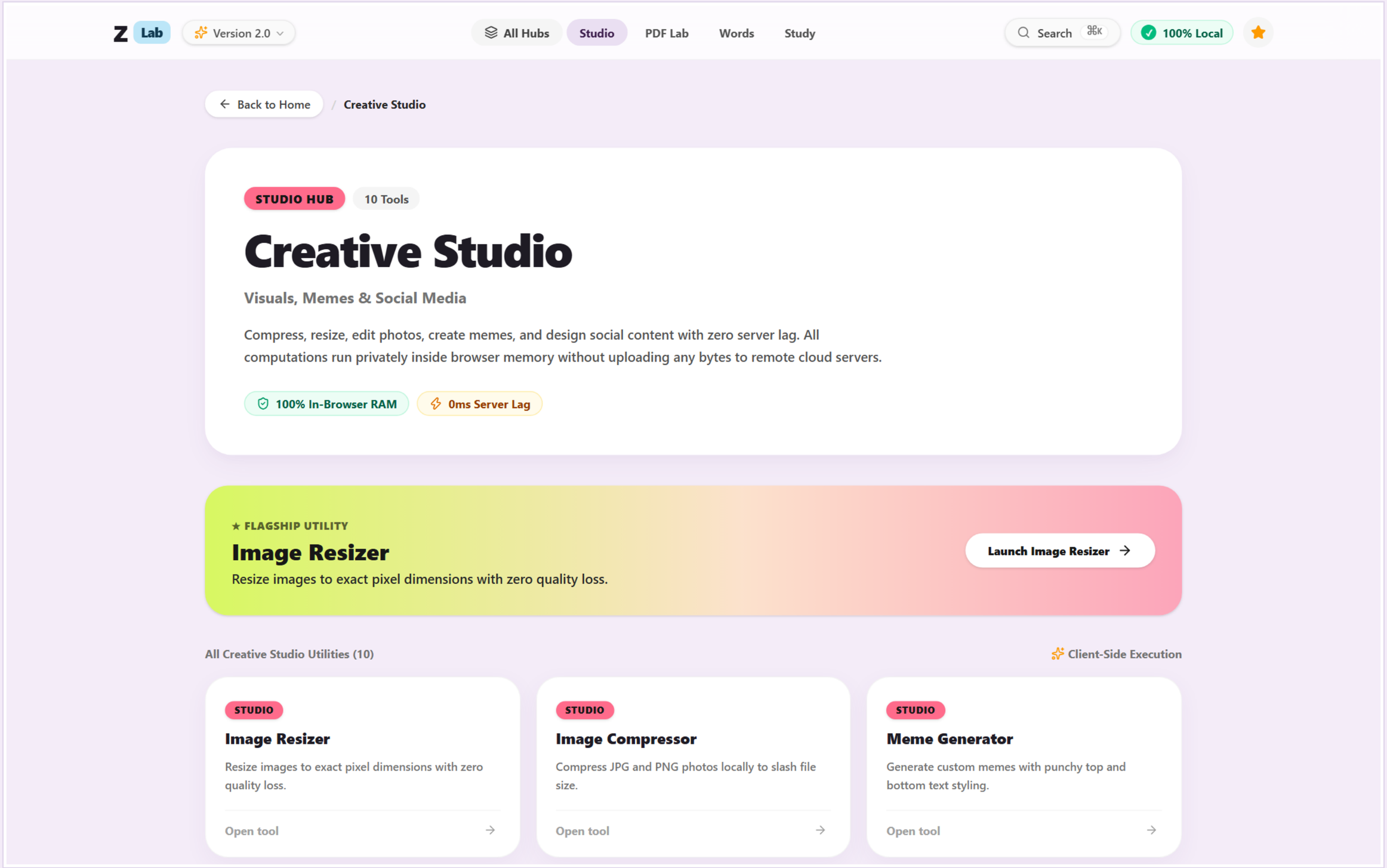
08. Category Hub -- Study & Focus Laboratory



DESKTOP VIEW (1440 x 900)

Route: /tools/creative-studio

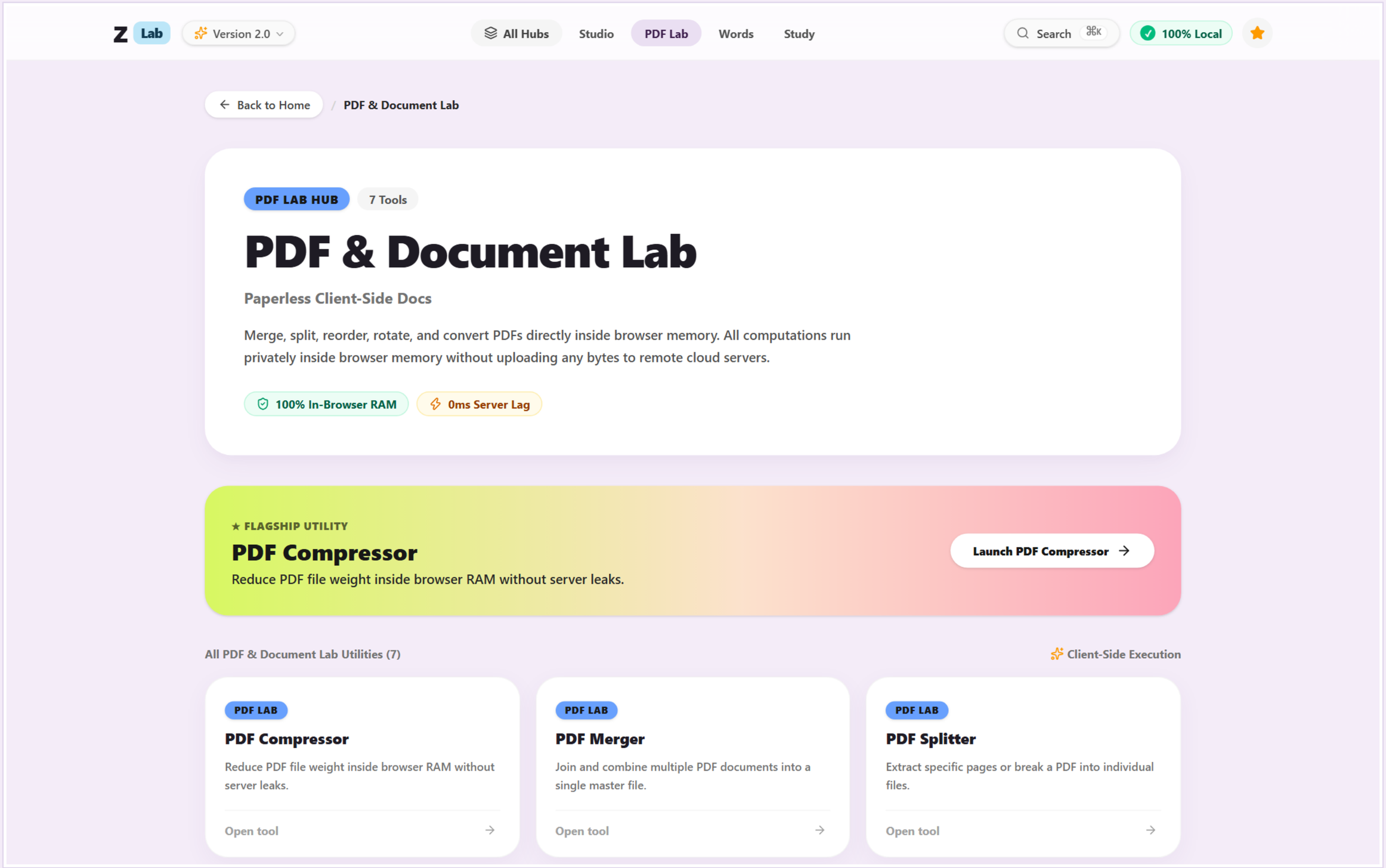
09. Category Hub -- Creative Studio Laboratory



DESKTOP VIEW (1440 x 900)

Route: /tools/pdf-document

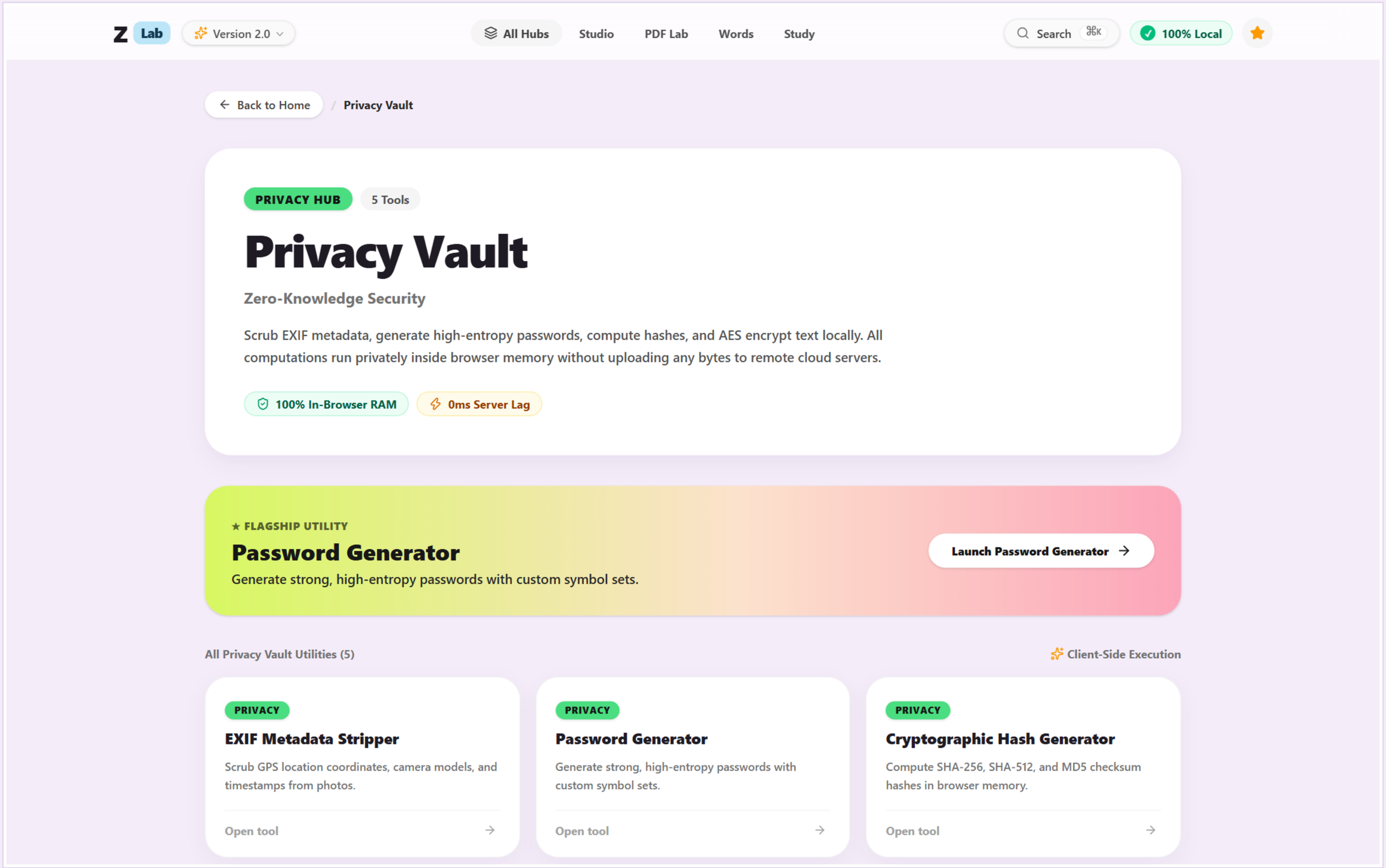
10. Category Hub -- PDF & Documents Laboratory



DESKTOP VIEW (1440 x 900)

Route: /tools/privacy-vault

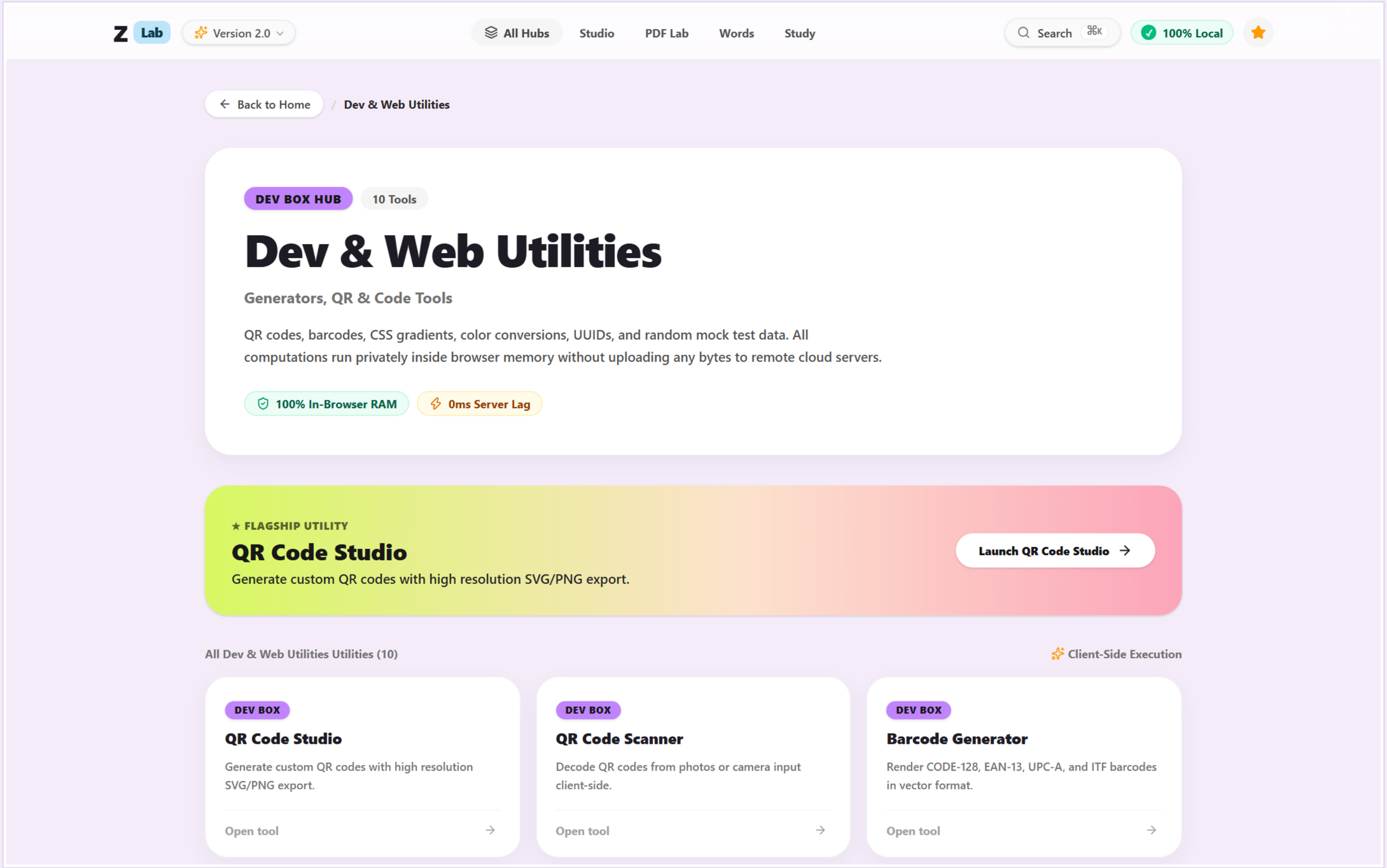
11. Category Hub -- Privacy & Security Vault



DESKTOP VIEW (1440 x 900)

Route: /tools/dev-utilities

12. Category Hub -- Developer Utilities Laboratory



DESKTOP VIEW (1440 x 900)

Route: /tools/text-writing

13. Category Hub -- Text & Writing Laboratory

Back to Home / Text & Writing Suite

WORDS HUB | 7 Tools

Text & Writing Suite

Clean Formatting & Metrics

Word counter, markdown live preview, case converter, citations, and instant text-to-speech. All computations run privately inside browser memory without uploading any bytes to remote cloud servers.

100% In-Browser RAM | 0ms Server Lag

★ FLAGSHIP UTILITY

Word & Reading Counter

Count words, characters, sentences, and estimated reading time live.

Launch Word & Reading Counter →

All Text & Writing Suite Utilities (7)

Client-Side Execution

WORDS

Word & Reading Counter

Count words, characters, sentences, and estimated reading time live.

Open tool →

WORDS

Character & Line Counter

Inspect detailed character metrics, line counts, byte sizes, and density.

Open tool →

WORDS

Case Converter

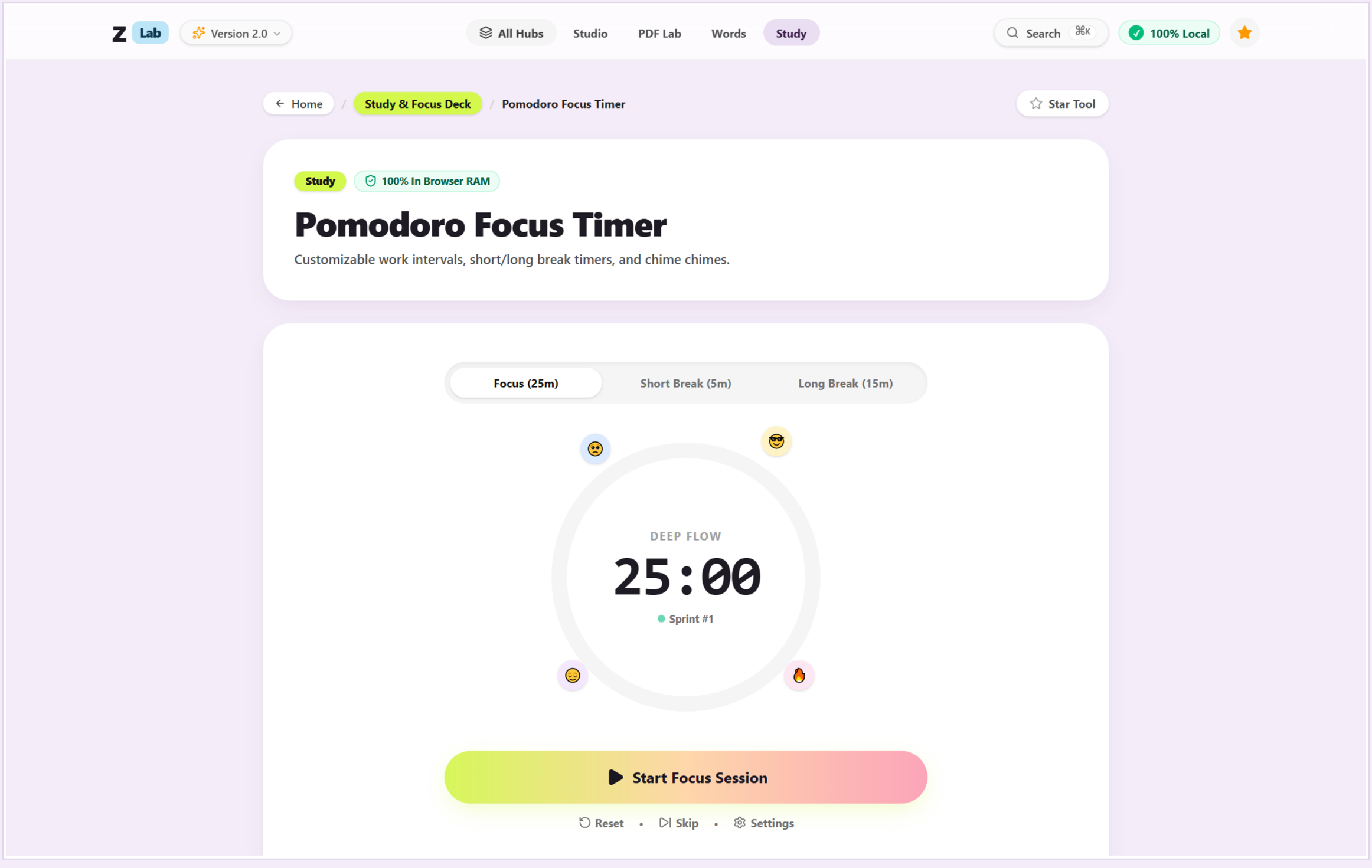
Transform text to UPPER, lower, Title Case, camelCase, and kebab-case.

Open tool →

DESKTOP VIEW (1440 x 900)

Route: /tools/study-focus/pomodoro-timer

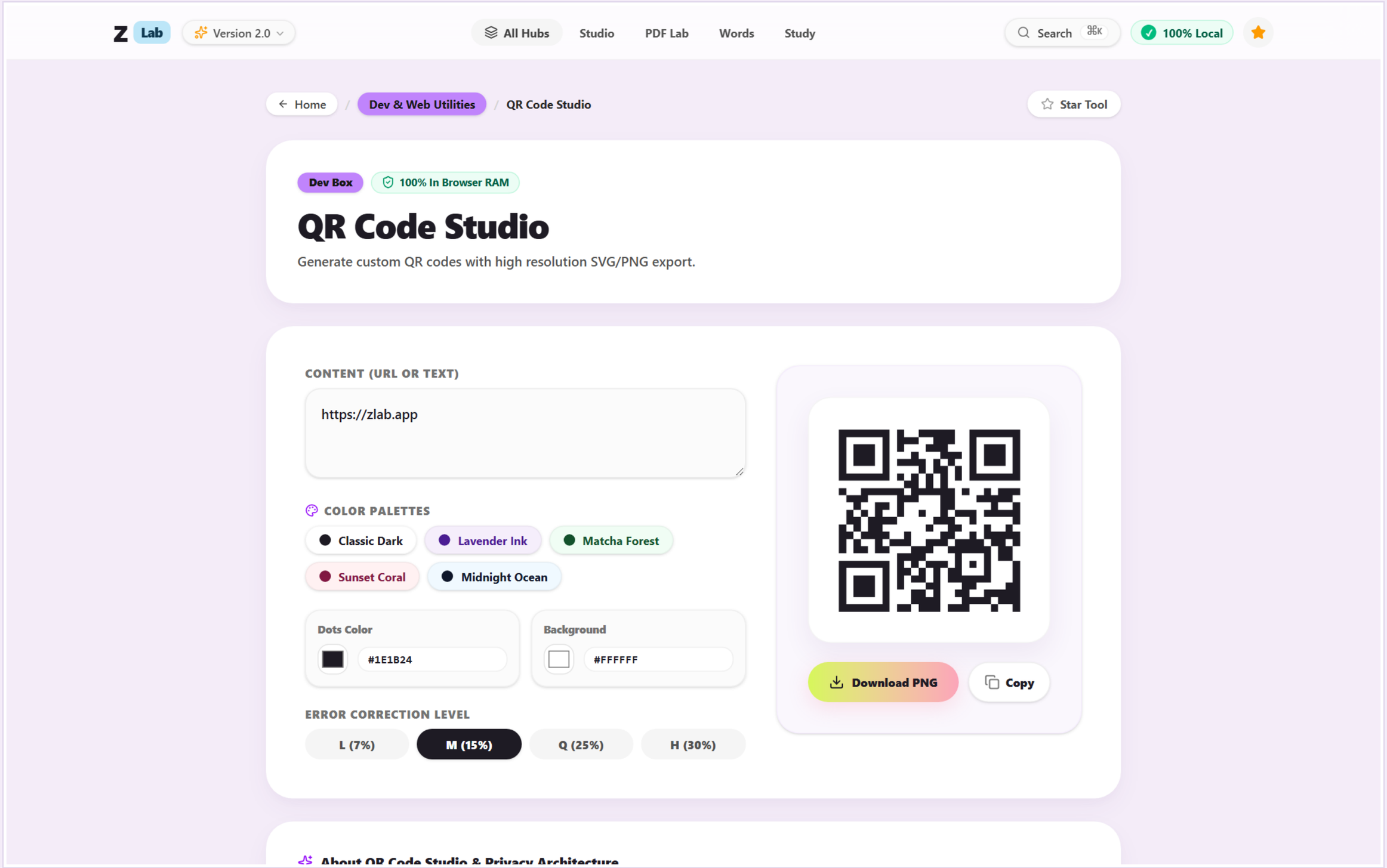
14. Flagship Tool -- MindMate Pomodoro Focus Session



DESKTOP VIEW (1440 x 900)

Route: /tools/dev-utilities/qr-generator

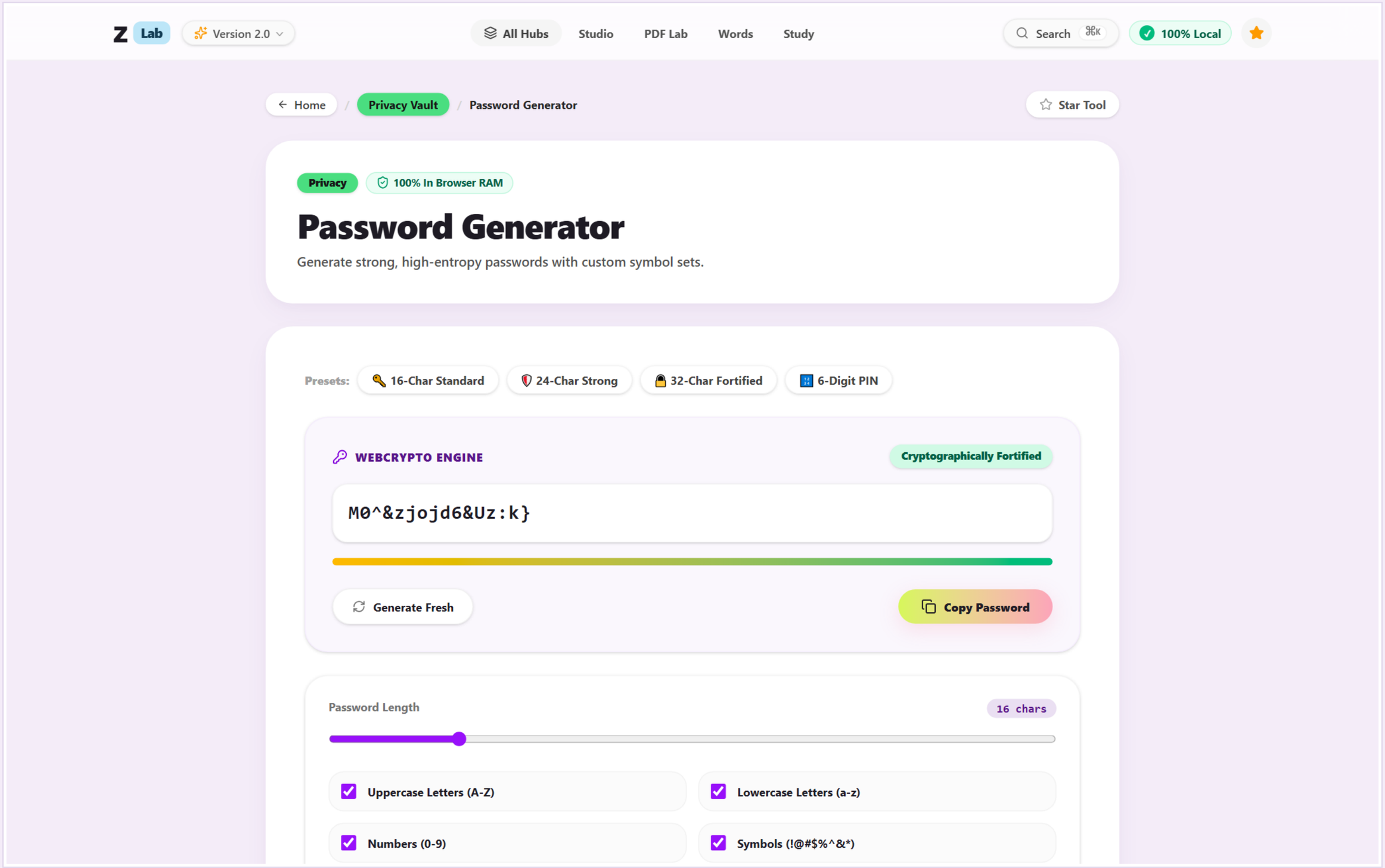
15. Flagship Tool -- QR Code Studio & Palette Customizer



DESKTOP VIEW (1440 x 900)

Route: /tools/privacy-vault/password-generator

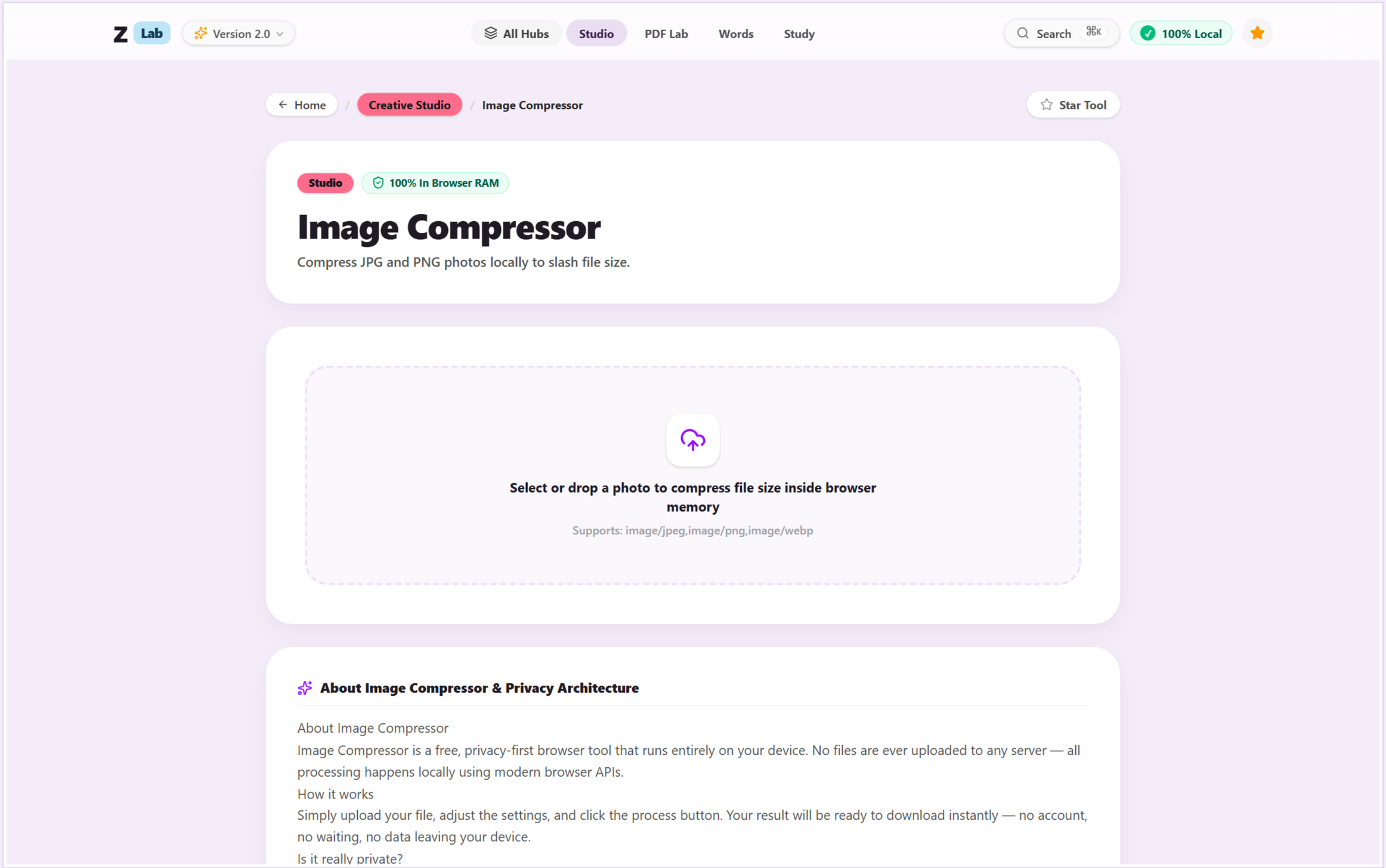
16. Flagship Tool -- WebCrypto Password Generator



DESKTOP VIEW (1440 x 900)

Route: /tools/creative-studio/image-compressor

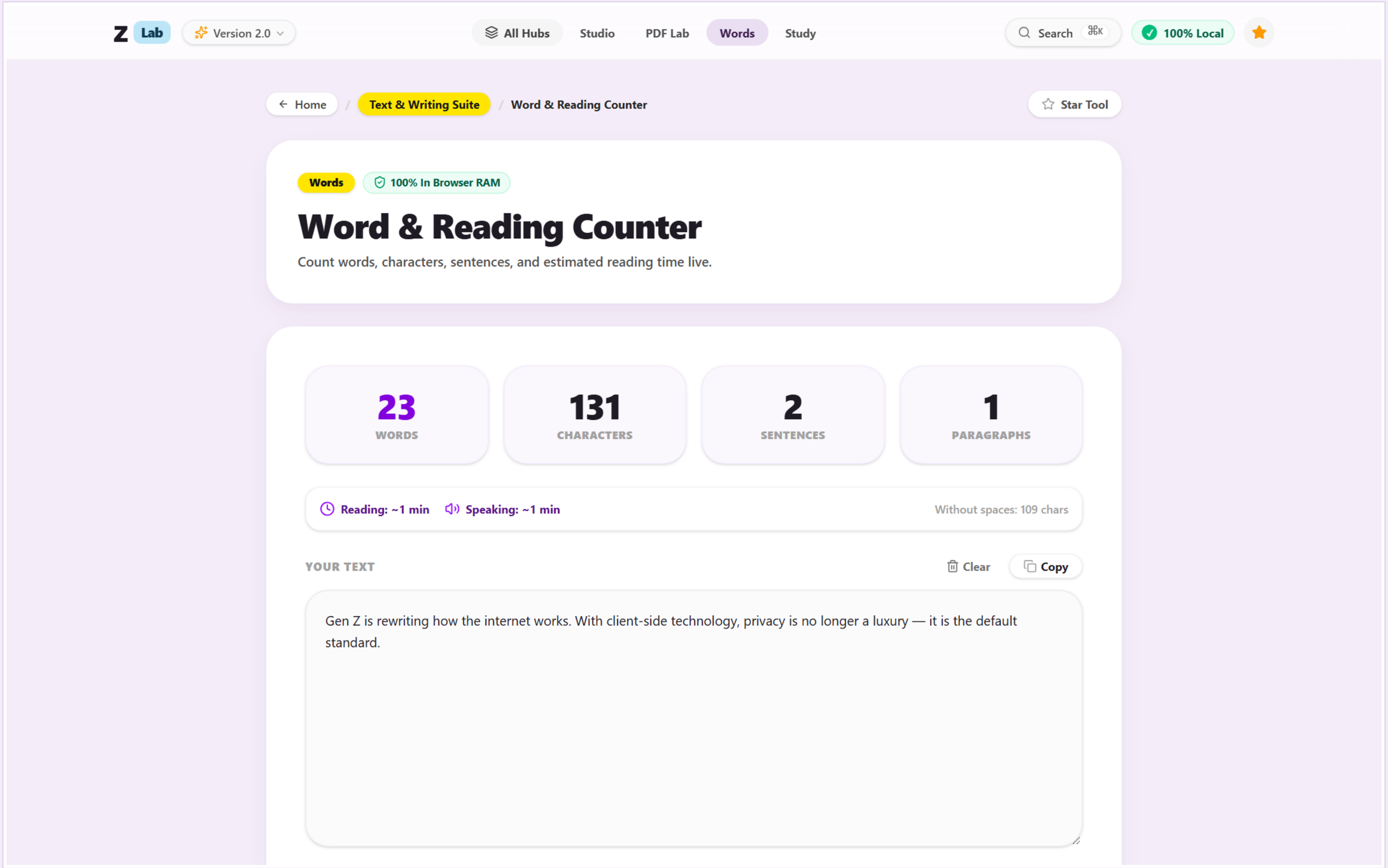
17. Flagship Tool -- In-Browser Image Compressor



DESKTOP VIEW (1440 x 900)

Route: /tools/text-writing/word-counter

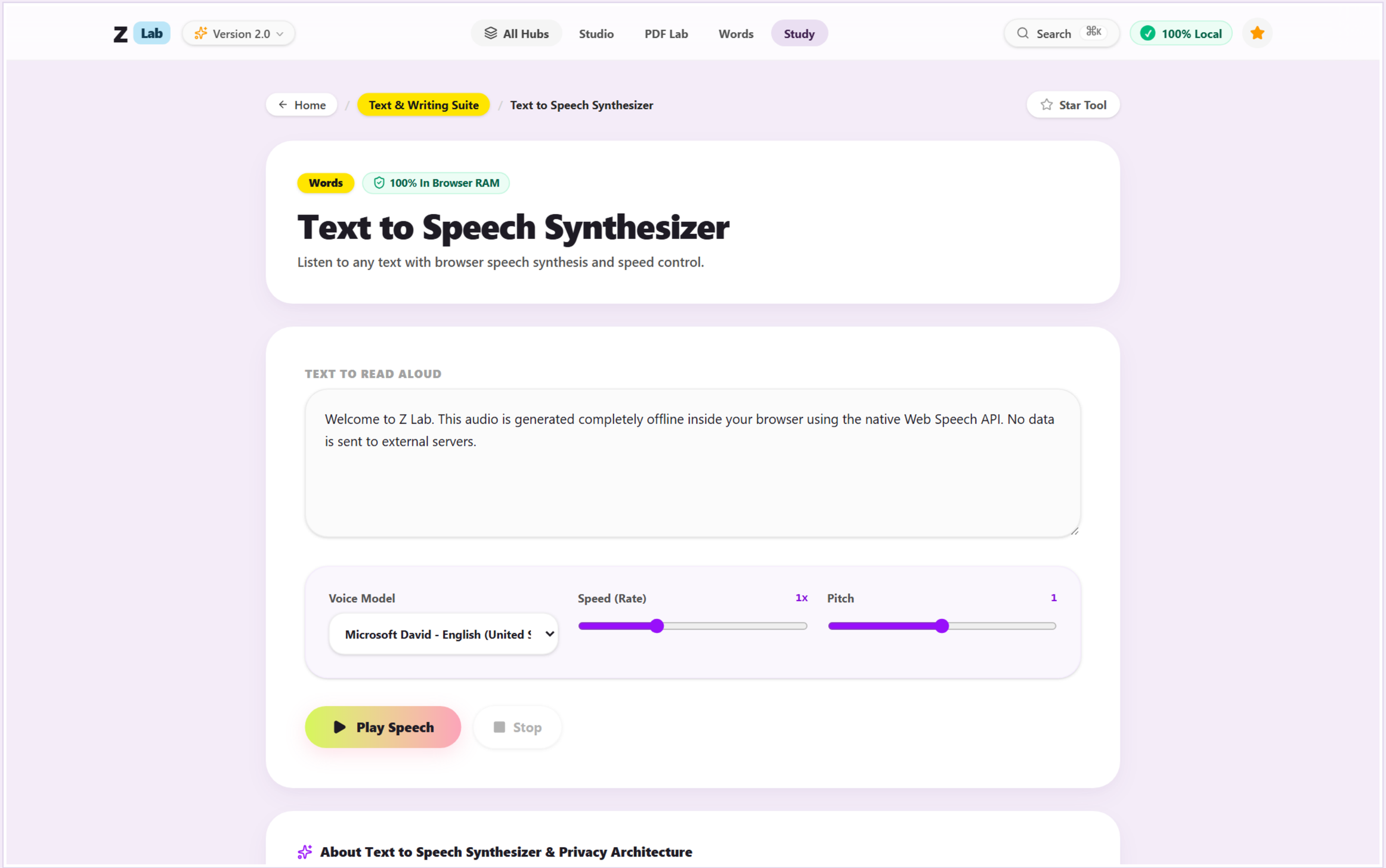
18. Flagship Tool -- Live Word & Reading Analytics



DESKTOP VIEW (1440 x 900)

Route: /tools/study-focus/text-to-speech

19. Flagship Tool -- Client-Side Speech Synthesis Engine



PART II

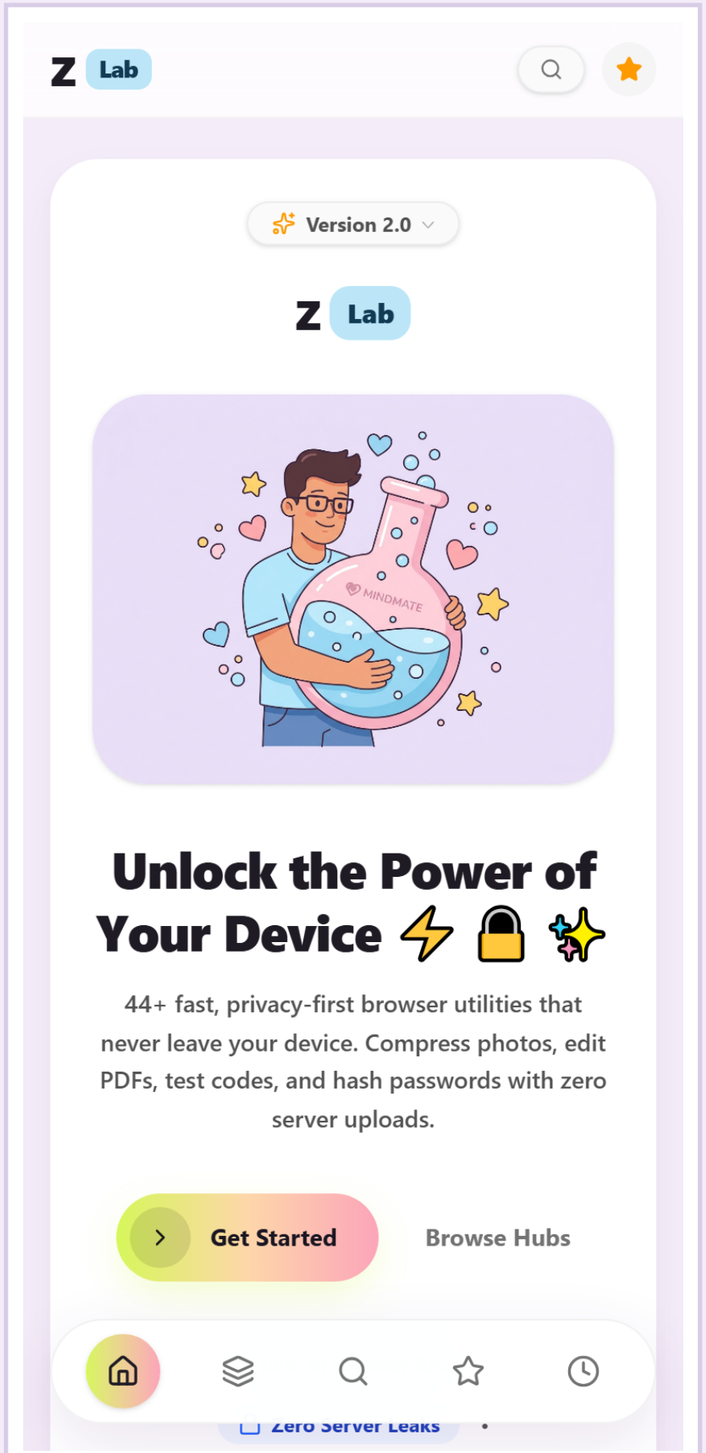
MOBILE ISLAND & TOUCH EXPERIENCE

390 x 844 Viewport - Floating Navigation Island & Mobile Drawers

MOBILE VIEW (390 x 844)

Route: /

01. Mobile Home Screen -- Onboarding Hero & Island Nav



MOBILE UX HIGHLIGHT

01. Mobile Home Screen -- Onboarding Hero & Island Nav

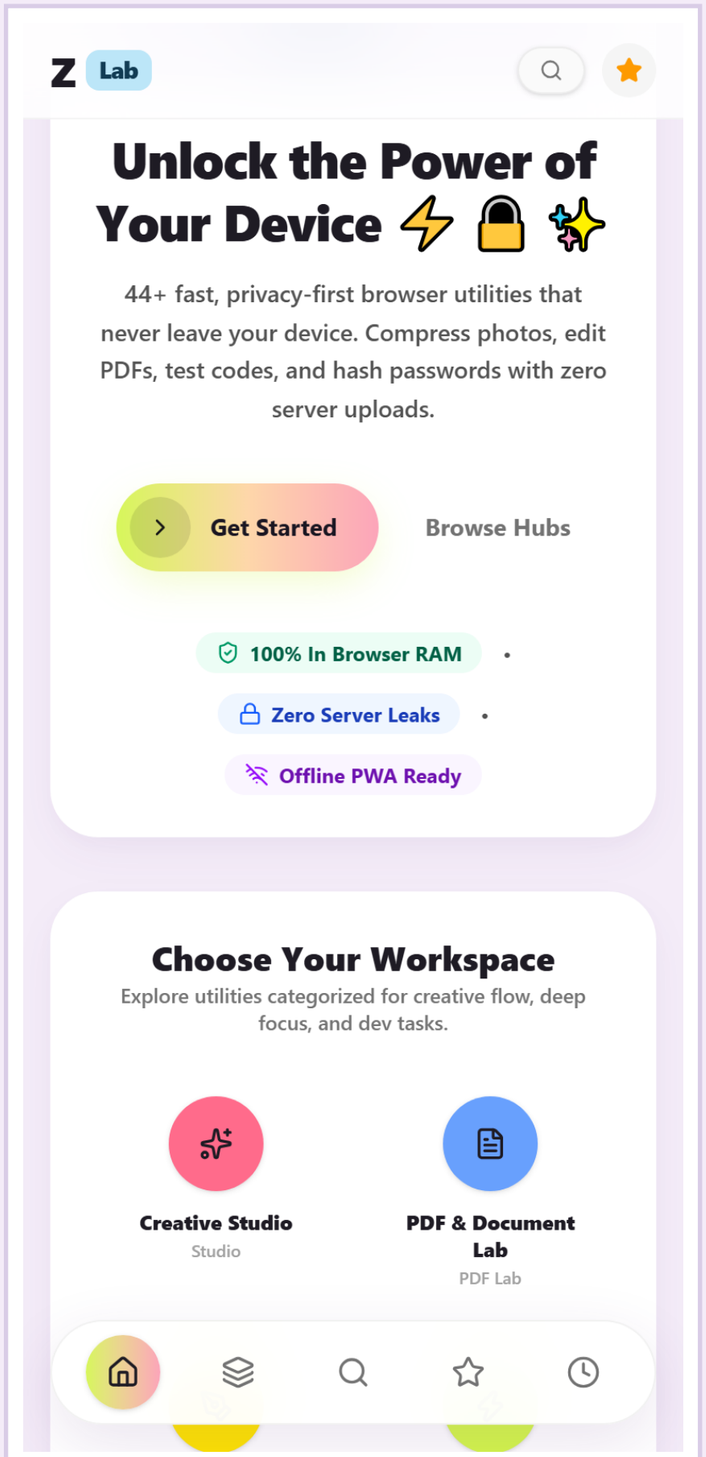
Touch-optimized header, character illustration, Version 2.0 chip, and floating pill bottom navigation.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /

02. Mobile Home Screen -- Dashboard & Category Grid



MOBILE UX HIGHLIGHT

02. Mobile Home Screen -- Dashboard & Category Grid

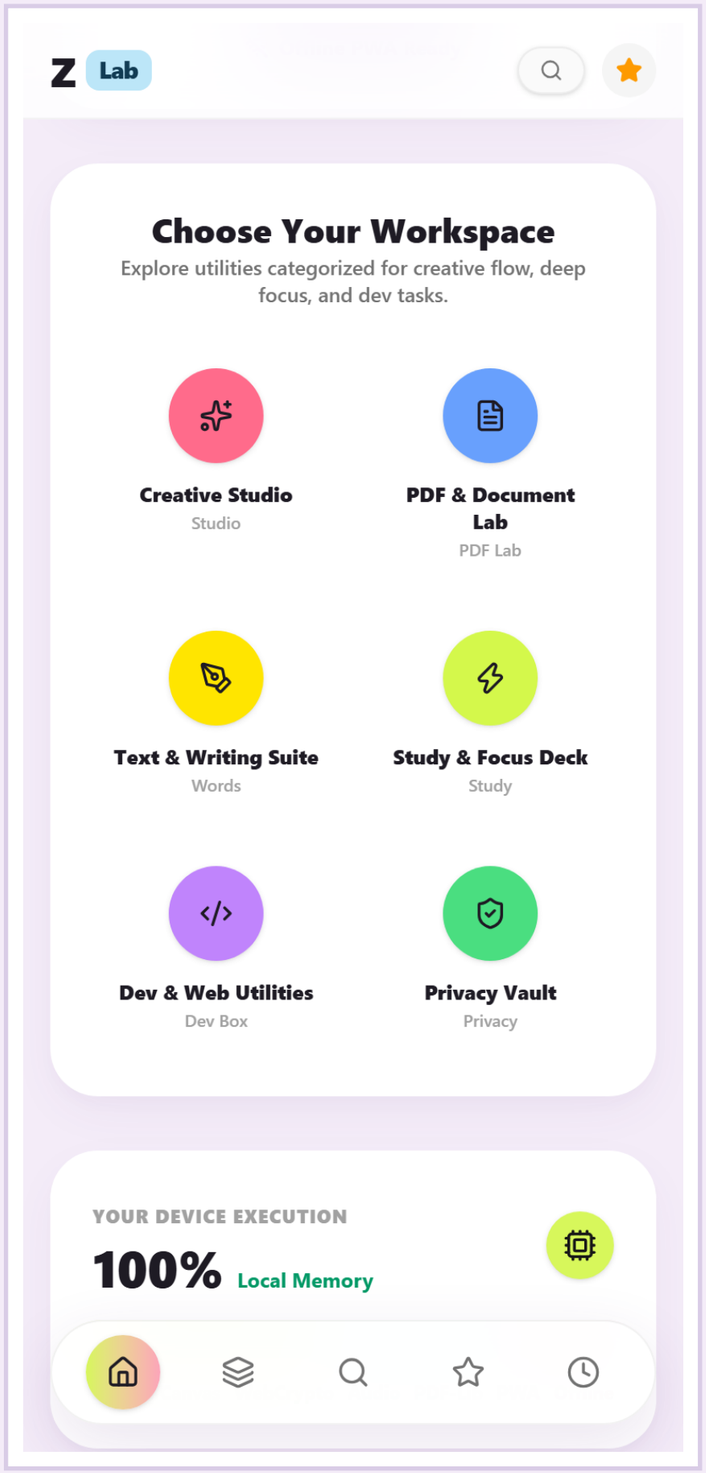
6 circular pastel category icon badges and device RAM execution meter stacked for mobile screens.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /

03. Mobile Home Screen -- Bento Action Widgets



MOBILE UX HIGHLIGHT

03. Mobile Home Screen -- Bento Action Widgets

Quick-access widgets optimized for vertical thumb-scrolling with instant copy and generator actions.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /

04. Mobile Home Screen -- Category Hubs Cards



MOBILE UX HIGHLIGHT

04. Mobile Home Screen -- Category Hubs Cards

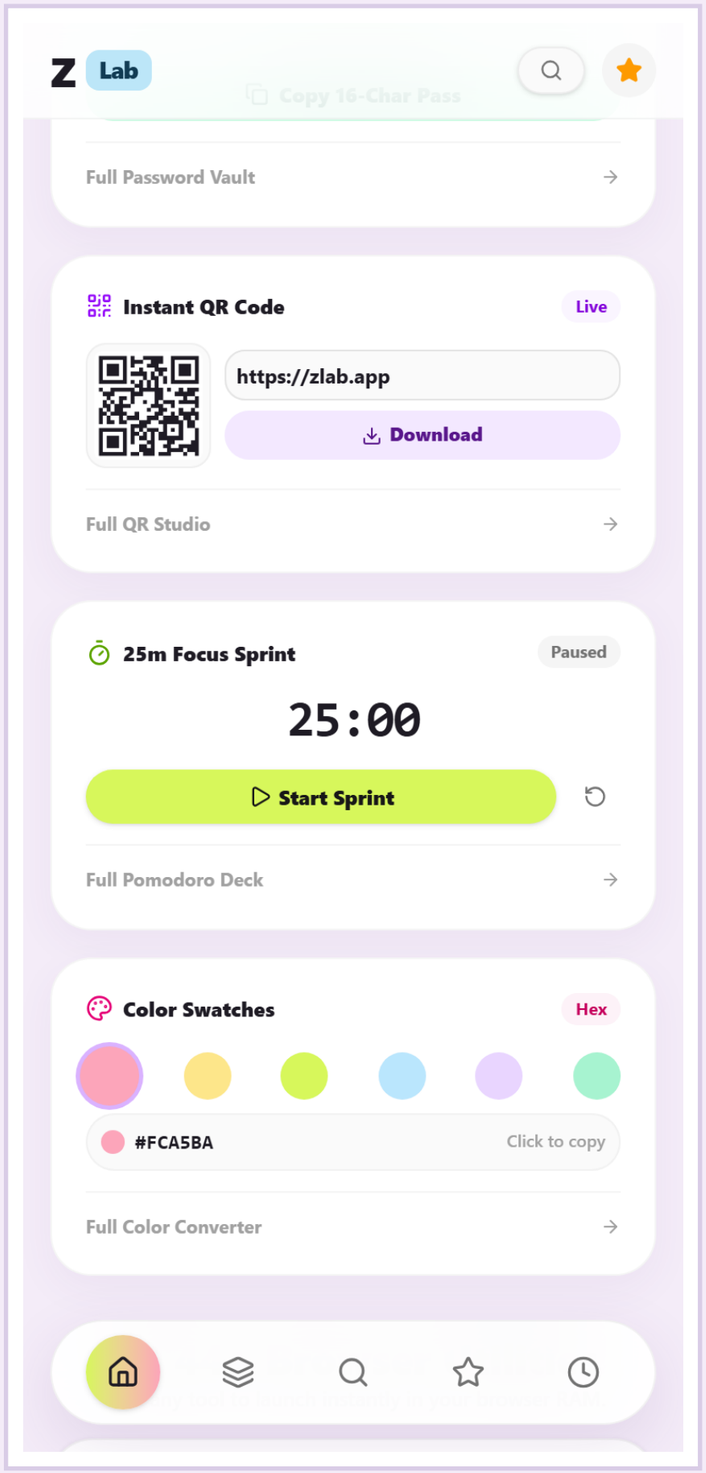
Elevated white cards with pure vector lab illustrations and smooth tap navigation.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /

05. Mobile Home Screen -- Tool Explorer & Filters



MOBILE UX HIGHLIGHT

05. Mobile Home Screen -- Tool Explorer & Filters

Horizontal swipeable category pills and card directory with favorite star triggers.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /

06. Mobile Omni-Search -- Fullscreen Command Palette



MOBILE UX HIGHLIGHT

06. Mobile Omni-Search -- Fullscreen Command Palette

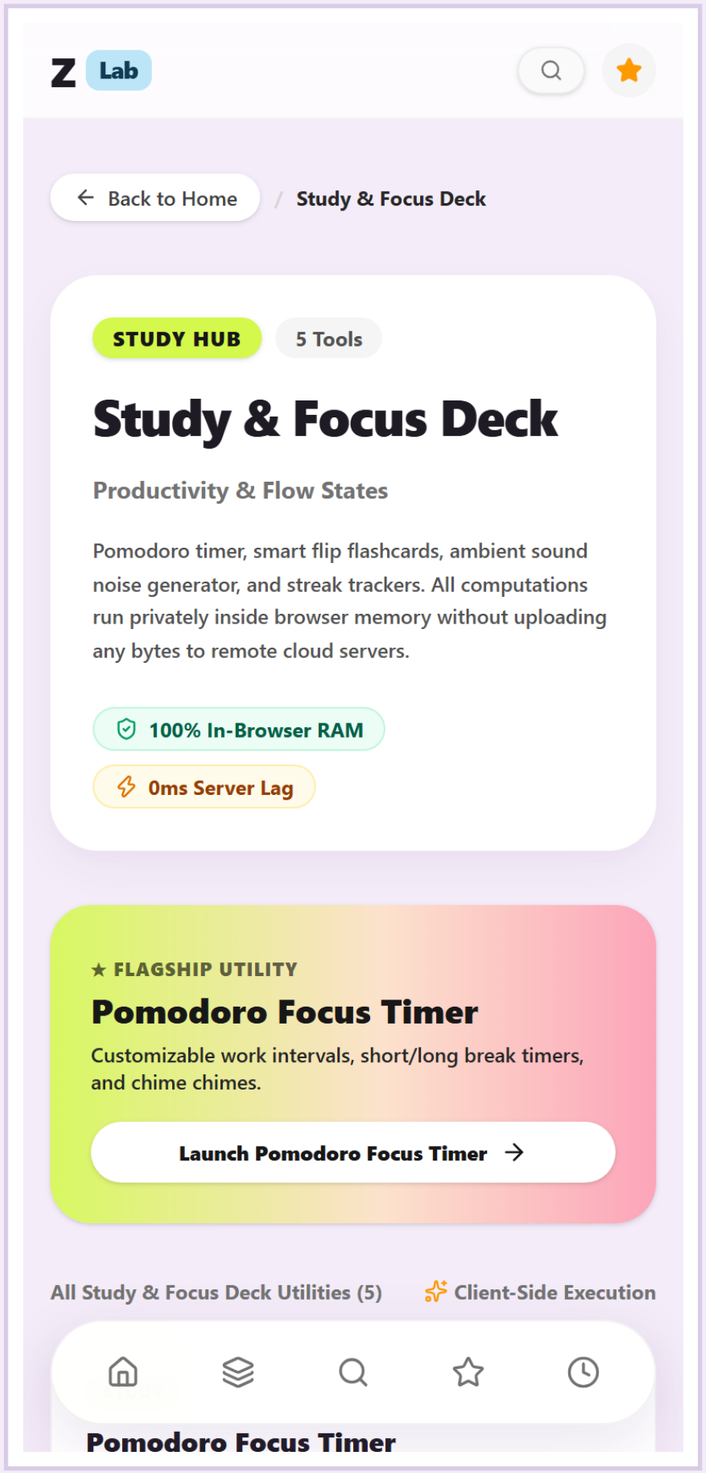
Triggered from the floating island bottom navigation bar for quick mobile tool discovery.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/study-focus

07. Mobile Category -- Study & Focus Hub



MOBILE UX HIGHLIGHT

07. Mobile Category -- Study & Focus Hub

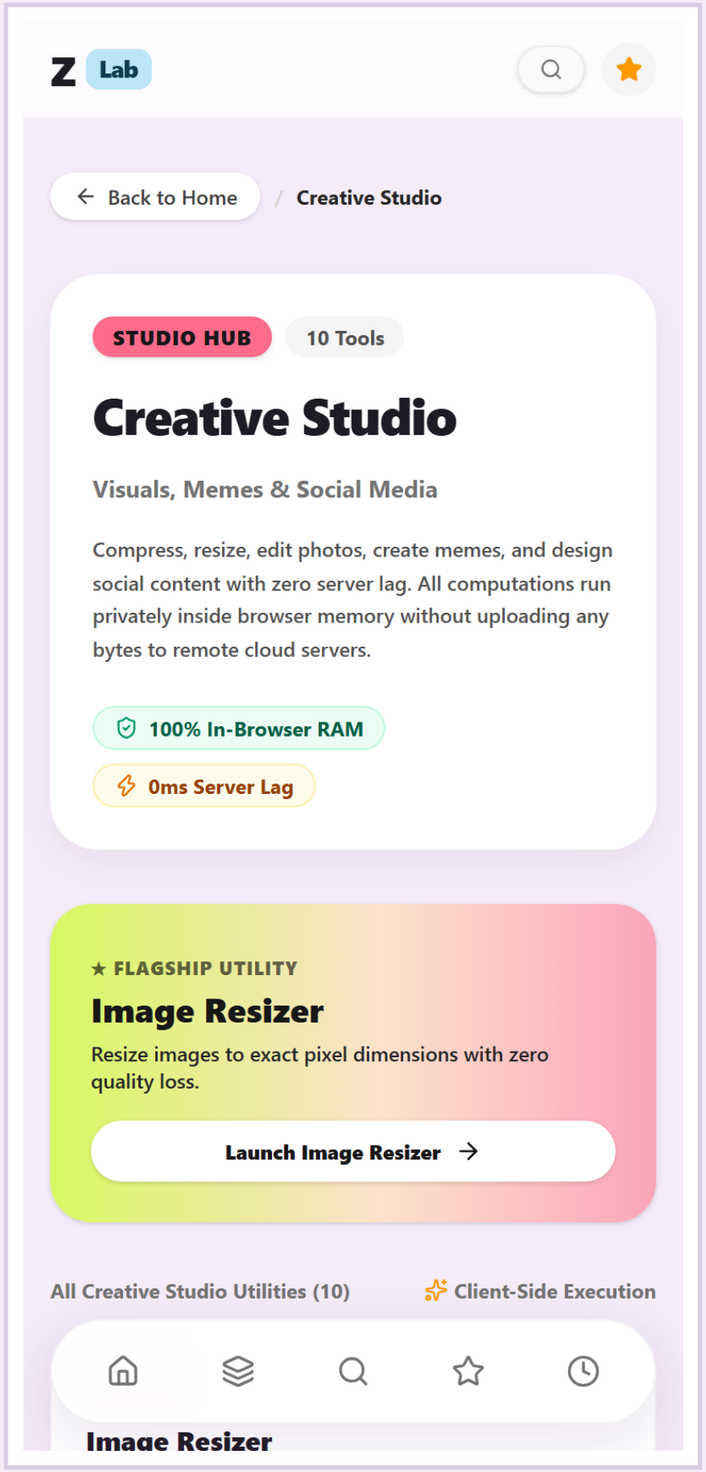
Full-bleed mobile category overview with sunset gradient flagship card and utility list.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/creative-studio

08. Mobile Category -- Creative Studio Hub



MOBILE UX HIGHLIGHT

08. Mobile Category -- Creative Studio Hub

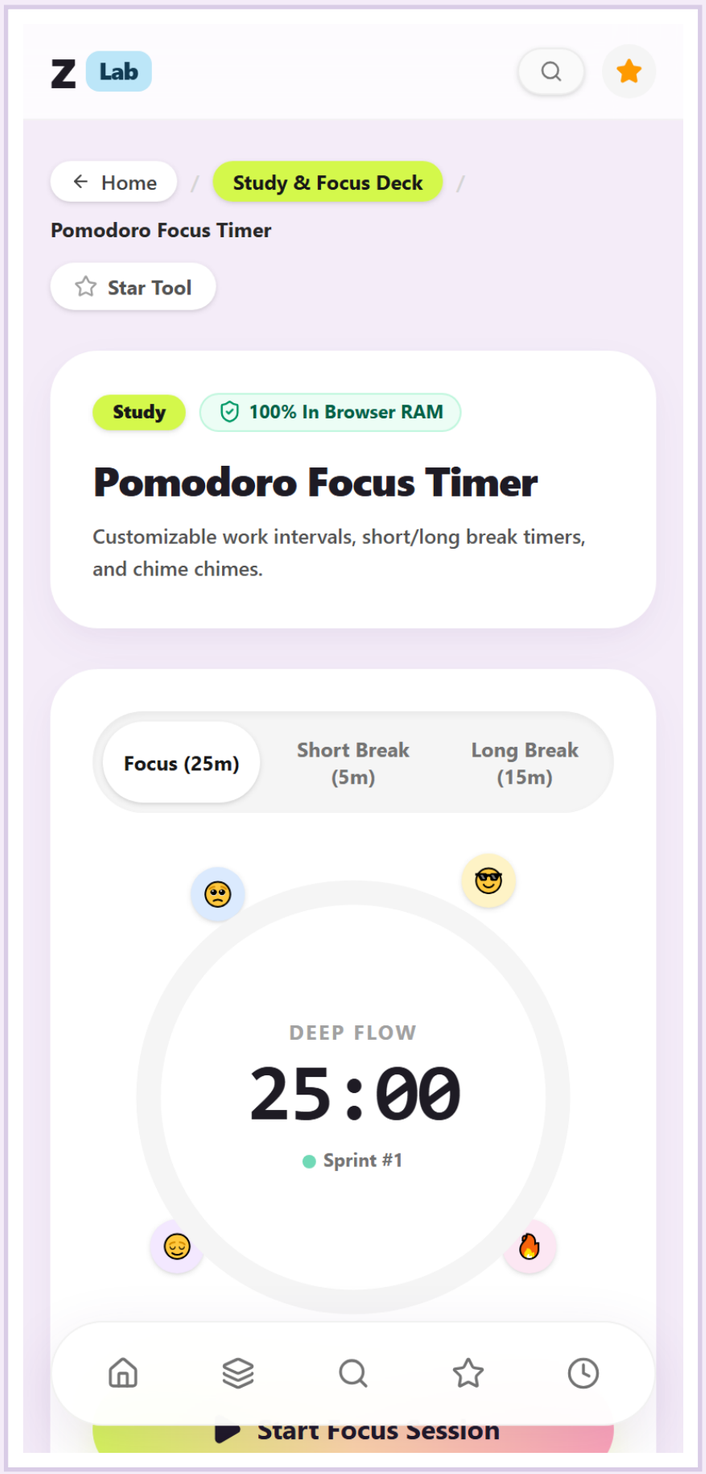
Mobile layout for image processing and canvas tools with zero upload privacy guarantee.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/study-focus/pomodoro-timer

09. Mobile Tool -- Pomodoro Focus Session



MOBILE UX HIGHLIGHT

09. Mobile Tool -- Pomodoro Focus Session

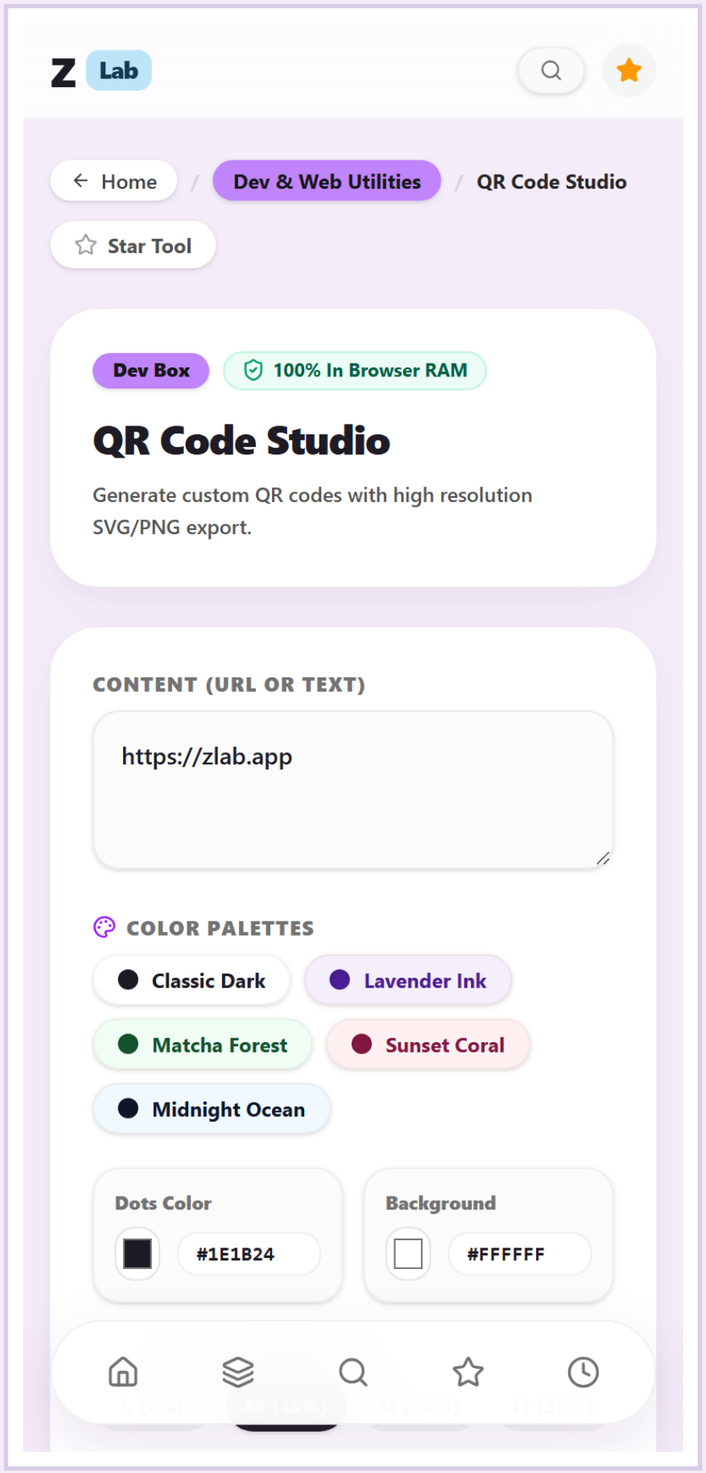
Circular radial timer dial with compact time chip matrix and mobile lo-fi controls.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/dev-utilities/qr-generator

10. Mobile Tool -- QR Code Studio



MOBILE UX HIGHLIGHT

10. Mobile Tool -- QR Code Studio

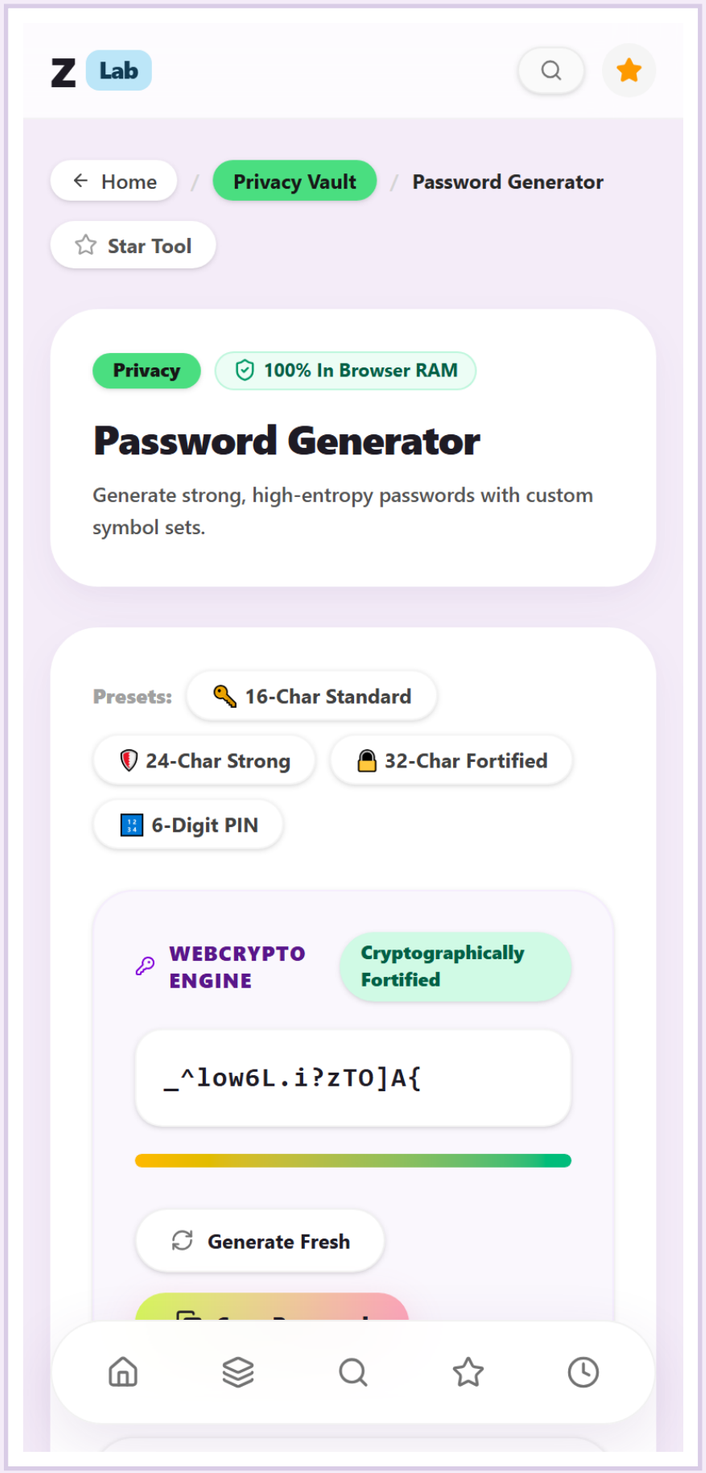
Vertical stacked layout: Content input, palette buttons, live QR preview, and download action.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/privacy-vault/password-generator

11. Mobile Tool -- WebCrypto Password Vault



MOBILE UX HIGHLIGHT

11. Mobile Tool -- WebCrypto Password Vault

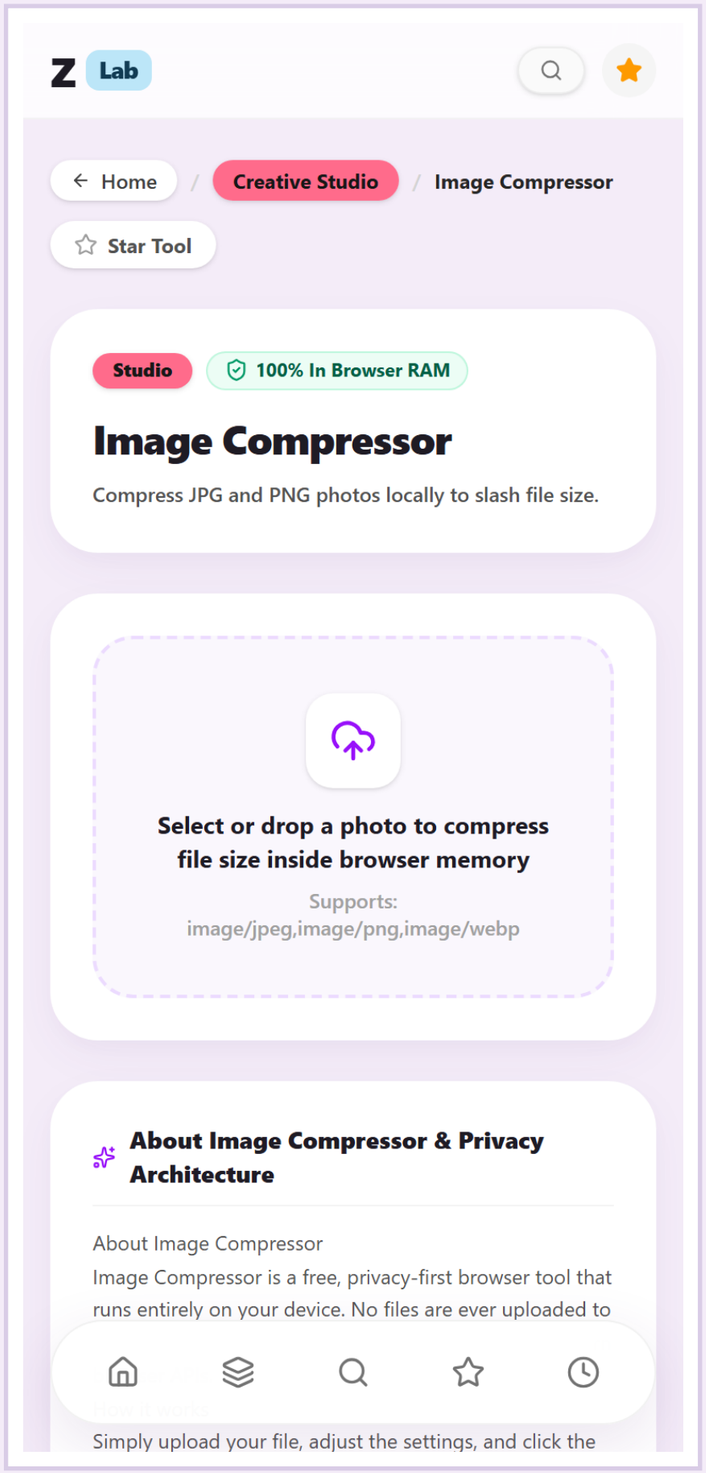
Mobile vault card with touch sliders, character switches, and gradient copy button.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/creative-studio/image-compressor

12. Mobile Tool -- In-Browser Image Compressor



MOBILE UX HIGHLIGHT

12. Mobile Tool -- In-Browser Image Compressor

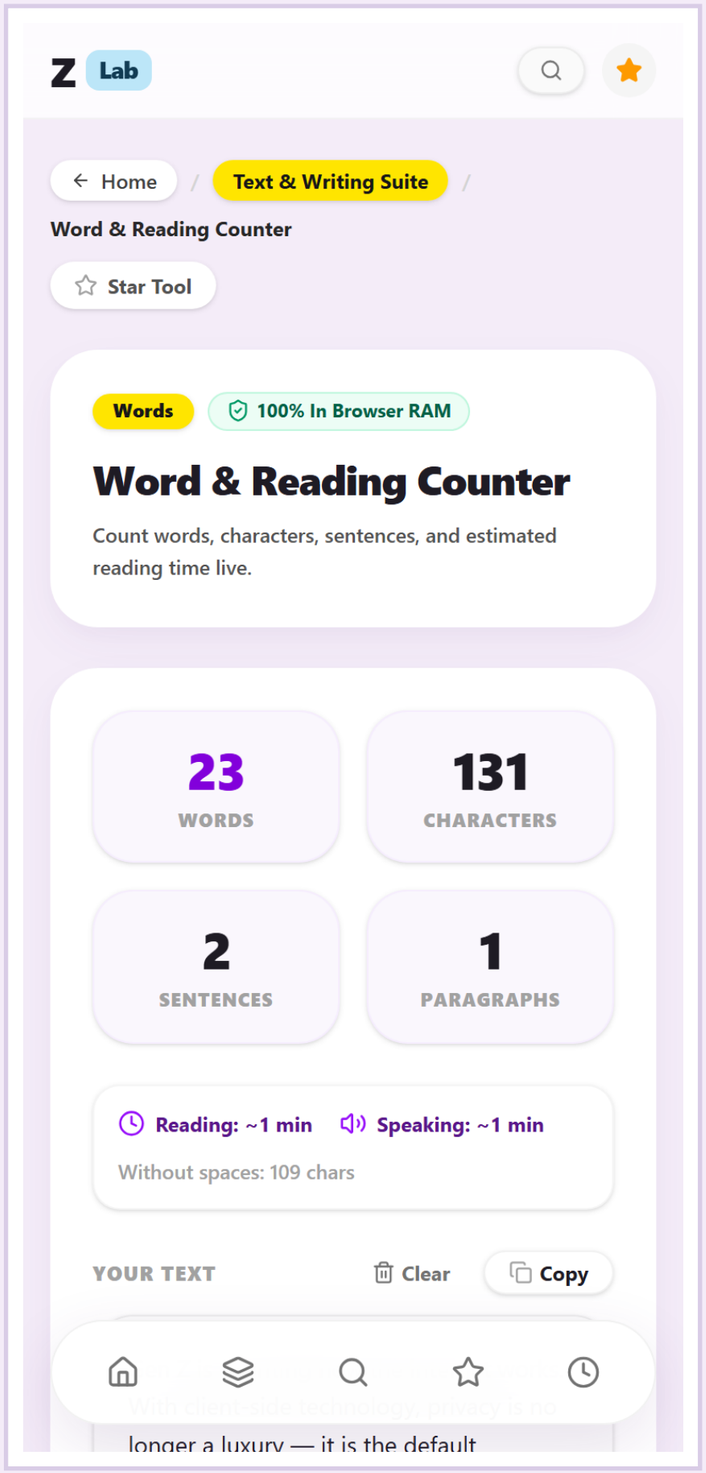
Mobile drag/tap file selection zone, quality slider, and responsive compression controls.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

MOBILE VIEW (390 x 844)

Route: /tools/text-writing/word-counter

13. Mobile Tool -- Word & Text Analytics



MOBILE UX HIGHLIGHT

13. Mobile Tool -- Word & Text Analytics

2x2 grid of pastel metric counters, reading stats, and full mobile typing workspace.

- Optimized for one-thumb reachability
- Floating island bottom navigation bar
- Client-side local execution without network latency

TECHNICAL SPECIFICATIONS

Z LAB Architecture & Feature Matrix

1. Creative Studio	**16 Tools**	HTML5 Canvas 2D API, Blob URL streaming, WebP & JPEG encoders
2. PDF & Documents	**6 Tools**	WebAssembly pdf-lib runtime, in-memory PDF page manipulation
3. Text & Writing	**8 Tools**	RegExp word frequency engines, Web Speech API synthesis
4. Study & Focus	**3 Tools**	Web Audio API ambient white noise oscillator, Lo-Fi synthesizer
5. Dev Utilities	**6 Tools**	QR Matrix math, Barcode canvas renderer, Base64 streaming
6. Privacy & Security	**5 Tools**	WebCrypto getRandomValues, SHA-256 / SHA-512 browser digest

Total Tools: 44 client-side utilities | Server Uploads: 0 bytes | Privacy Guarantee: 100% In-RAM